

Outline of the lecture

- **COMPLEMENT OF A NUMBER**
 - **DECIMAL NUMBER SYSTEM**
 - ✓ 9's complement
 - ✓ 10's complement
 - ✓ Subtraction using 9's complement
 - ✓ Subtraction using 10's complement
 - **BINARY NUMBER SYSTEM**
 - ✓ 1's complement
 - ✓ 2's complement
 - ✓ Subtraction using 1's complement
 - ✓ Subtraction using 2's complement
 - **'r' NUMBER SYSTEM**
 - ✓ $(r-1)$'s complement
 - ✓ (r) 's complement
 - ✓ Subtraction using $(r-1)$'s complement
 - ✓ Subtraction using (r) 's complement
- **COMPARISONS**
- **REPRESENTATION OF SIGNED NUMBERS**
- **PROBLEMS FOR PRACTICE**

NEED FOR COMPLEMENTS

- ▶ For any radix 'r' number system, there exists two complements:
 - ▶ (r-1) complement
 - ▶ (r) complement
- ▶ Complements help in converting the arithmetic subtraction operation into arithmetic addition operation.
- ▶ Hence, all arithmetic operations i.e. addition, subtraction, multiplication and division can be performed using only adders.

DECIMAL NUMBER SYSTEM

► 9's Complement

► Is obtained by subtracting each digit in the number from 9

e.g. 9's complement of 4397 = $9999 - 4397 = 5602$

e.g. 9's complement of 678.42 = $999.99 - 678.42 = 321.57$

	9	9	9	9	
-	4	3	9	7	Number
	5	6	0	2	9's complement

	9	9	9	.	9	9	
-	6	7	8	.	4	2	Number
	3	2	1	.	5	7	9's complement

► 10's Complement

► Is obtained by adding '1' to right most digit of 9's complement of the number

e.g. 10's complement of 4397 = $(9999 - 4397) + 1 = 5603$

e.g. 10's complement of 678.42 = $(999.99 - 678.42) + 1 = 321.58$

	9	9	9	9	
-	4	3	9	7	Number
	5	6	0	2	9's complement
+				1	
	5	6	0	3	10's complement

	9	9	9	.	9	9	
-	6	7	8	.	4	2	Number
	3	2	1	.	5	7	9's complement
+						1	
	3	2	1	.	5	8	10's complement

DECIMAL NUMBER SYSTEM

► Subtraction using 9's Complement

► Smaller decimal number from a larger one

- add the 9's complement of the subtrahend (number to be subtracted) to the minuend
- add the carry generated to the obtained sum

e.g. $99 - 83 = 99 + (\text{9's complement of } 83)$

	9	9	Minuend
+	1	6	9's complement of 83
	1	5	Sum
+		1	Carry
	1	6	

Carry

Result

DECIMAL NUMBER SYSTEM

► Subtraction using 9's Complement

► Larger decimal number from a smaller one

- add the 9's complement of the subtrahend (number to be subtracted) to the minuend
- no carry is generated
- the result is negative with magnitude in 9's complement form

e.g. $34 - 49 = 34 + (\text{9's complement of } 49)$

	3	4	Minuend
+	5	0	9's complement of 49
	8	4	Sum
-	1	5	9's complement of 84

No Carry

Result

DECIMAL NUMBER SYSTEM

► Subtraction using 10's Complement

► Smaller decimal number from a larger one

- add the 10's complement of the subtrahend (number to be subtracted) to the minuend
- discard the carry generated

e.g. $54 - 36 = 54 + (10\text{'s complement of } 36)$

	5	4	Minuend
+	6	4	10's complement of 36
Carry discarded	1	1	8
Result	1	8	

DECIMAL NUMBER SYSTEM

► Subtraction using 10's Complement

► Larger decimal number from a smaller one

- add the 10's complement of the subtrahend (number to be subtracted) to the minuend
- no carry is generated
- the result is negative with magnitude in 10's complement form.

e.g. $28 - 63 = 28 + (\text{10's complement of } 63)$

	2	8	Minuend
+	3	7	10's complement of 63
	6	5	Sum
	-	3	5
			10's complement of 65

No Carry

Result

BINARY NUMBER SYSTEM

► 1's Complement

- Is obtained by subtracting each digit in the number from 1
- OR change all 1's to 0 and all 0's to 1

e.g. 1's complement of 11010 = $11111 - 11010 = 00101$

► 2's Complement

- Is obtained by adding '1' to right most digit of 1's complement of the number

e.g. 2's complement of 1001 = 1's complement + 1
= $(1111 - 1001) + 1$
= $0110 + 1$
= 0111

BINARY NUMBER SYSTEM

► Subtraction using 1's Complement

► Smaller binary number from a larger one

- add the 1's complement of the subtrahend (number to be subtracted) to the minuend
- add the carry generated to the obtained sum. This is referred as END-AROUND-CARRY

e.g. $11101 - 10101 = 11101 + (1\text{'s complement of } 10101)$

	1	1	1	0	1	Minuend
+	0	1	0	1	0	1's complement of 10101
Carry	1	0	0	1	1	Sum
+					1	Carry
Result	0	1	0	0	0	

BINARY NUMBER SYSTEM

► Subtraction using 1's Complement

► Larger binary number from a smaller one

- add the 1's complement of the subtrahend (number to be subtracted) to the minuend
- no carry is generated
- the result is negative with magnitude in 1's complement form.

e.g. $1000 - 1010 = 1000 + (1\text{'s complement of } 1010)$

	1	0	0	0	Minuend
+	0	1	0	1	1's complement of 1010
No Carry	1	1	0	1	Sum
Result	-	0	0	1	1's complement of 1101

BINARY NUMBER SYSTEM

► Subtraction using 2's Complement

► Smaller binary number from a larger one

- add the 2's complement of the subtrahend (number to be subtracted) to the minuend
- discard the carry generated (there is always in this case)

e.g. $1100 - 1001 = 1100 + (\text{2's complement of } 1001)$

	1	1	0	0	Minuend
+	0	1	1	1	2's complement of 1001
	1	0	0	1	Sum
	0	0	1	1	

Carry discarded

Result

BINARY NUMBER SYSTEM

► Subtraction using 2's Complement

► Larger binary number from a smaller one

- add the 2's complement of the subtrahend (number to be subtracted) to the minuend
- no carry is generated
- the result is negative with magnitude in 2's complement form

e.g. $1000 - 1100 = 1000 + (\text{2's complement of } 1100)$

	1	0	0	0	Minuend
+	0	1	0	0	2's complement of 1100
No Carry	1	1	0	0	Sum
Result	-	0	1	0	2's complement of 1100

RADIX 'r' NUMBER SYSTEM

► $(r-1)^{\text{th}}$ Complement

► $(r-1)^{\text{th}} = r^n - r^{-m} - N$

where, r is the radix or base

n is number of maximum digits before decimal

m is number of maximum digits after decimal

N is the number to be subtracted

e.g. $(201)_{10} - (101)_{10}$ using $(r-1)^{\text{th}}$ complement,

i.e. $(10-1)^{\text{th}} = 9^{\text{th}}$ complement

For $(101)_{10}$, 9^{th} complement $= 10^3 - 10^{-0} - 101$

$$= 1000 - 1 - 101$$

$$= 1000 - 102 = 898$$

For subtraction: $201 + 898 = 1099$,

add carry, result is 100

	2	0	1	Minuend
+	8	9	8	9's complement of 101
	1	0	9	Sum
+			1	Carry
	1	0	0	Result

RADIX 'r' NUMBER SYSTEM

► r^{th} Complement

► $r^{\text{th}} = r^n - N$

where, r is the radix or base

n is number of maximum digits before decimal

N is the number to be subtracted

e.g. $(2365)_{10} - (3625)_{10}$ using r^{th} complement,

i.e. 10^{th} complement

For $(3625)_{10}$, 10^{th} complement = $10^4 - 3625$

$$= 10000 - 3625$$

$$= 6375$$

For subtraction: $2365 + 6375 = 8740$

No carry, result is (-1260)

No Carry

Result

	2	3	6	5	Minuend
+	6	3	7	5	10's complement of 3625
	8	7	4	0	Sum
-	1	2	6	0	10's complement of 8740

COMPARISON

1'S COMPLEMENT

- ▶ Easily obtained using an inverter
- ▶ Used in logical manipulation for inversion operation
- ▶ Requires 2 operations:
 - Logical operation
 - End around carry

2'S COMPLEMENT

- ▶ Obtained by adding one unit to 1's complement
- ▶ Used for arithmetic applications such as adders
- ▶ Requires 1 operations:
 - Logical operation

REPRESENTATION OF SIGNED NUMBERS

- ▶ In all number systems, both positive and negative numbers are possible
 - ▶ Positive numbers represented by ‘+’
 - ▶ Negative numbers represented by ‘-’
- ▶ Negative Binary numbers are represented by
 - ▶ Sign Magnitude representation
 - ▶ 1’s complement representation
 - ▶ 2’s complement representation

Problems

1. $(18 - 6)$ using 9's complement
2. $(21 - 38)$ using 9's complement
3. $(347 - 265)$ using 10's complement
4. $(74 - 85)$ using 10's complement
5. $(1111 - 1010)$ using 1's complement
6. $(1000 - 1100)$ using 1's complement
7. $(1111 - 1010)$ using 2's complement
8. $(1000 - 1010)$ using 2's complement
9. $(1011.01)_2 - (1100.00)_2$ using $(r-1)^{\text{th}}$ complement
10. Represent $(-31)_{10}$ and $(-99)_{10}$ in
 - a) 8-bit binary signed magnitude representation.
 - b) 2's complement representation
 - c) 1's complement representation